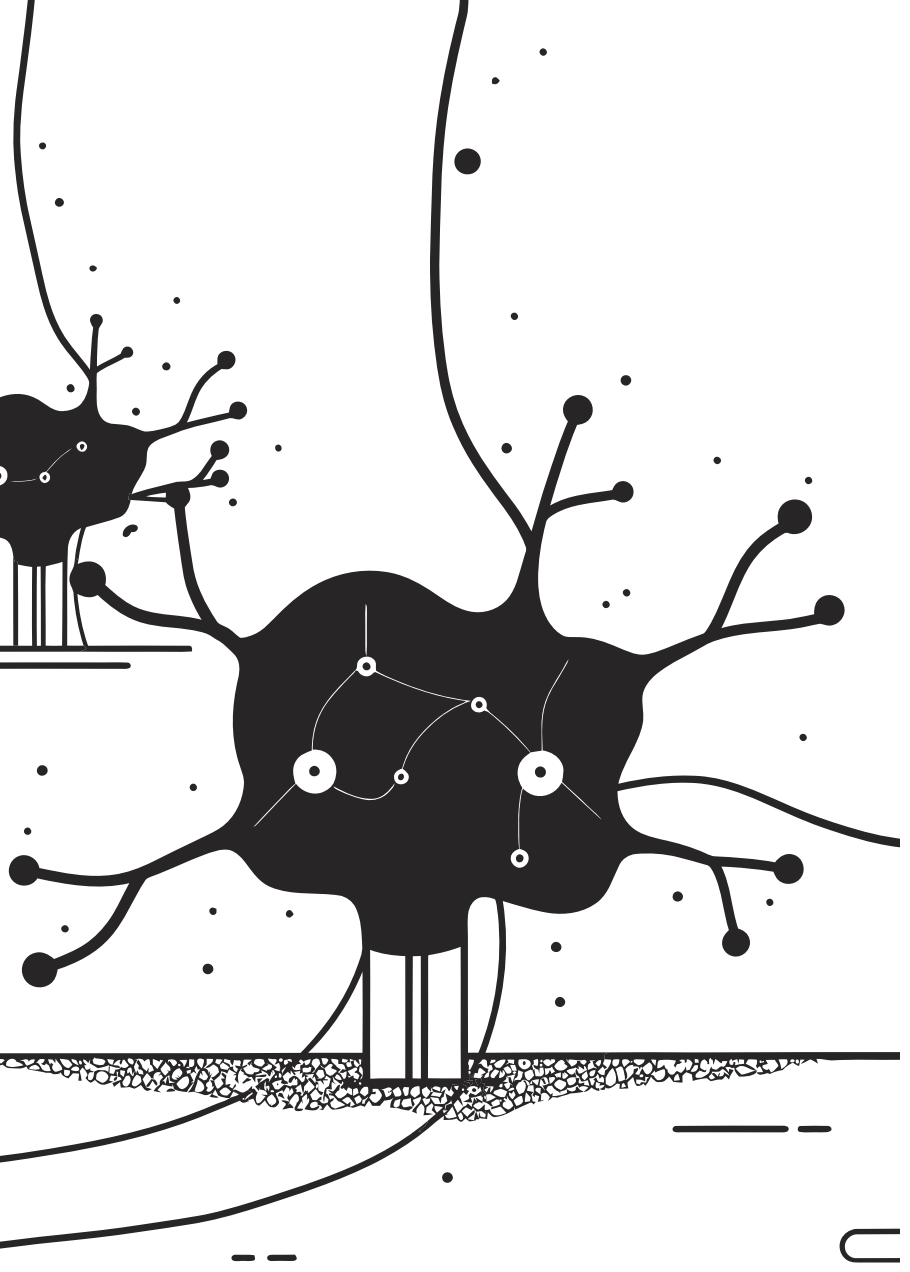


PDPL Keystone Articles and the Immutability Tension

Where an immutable ledger clashes with the right to erasure - and how to mitigate it.

Source: Finck (2019) STOA PE 634.445, sections 3-4, paired with the primary PDPL Article text.

The PDPL Keystone Articles

1

Article 12 - Cross-border Transfers

Personal data may not leave Bahrain unless the destination is on the Authority's adequacy list, or another lawful basis applies.

2

Article 22 - Automated Decisions

A person can object to a decision made solely by automated processing - such as credit scoring - and request human reconsideration.

3

Article 23 - Rectification and Erasure

A person can ask the controller to rectify, block or erase their personal data. The controller must respond free of charge within the statutory timeframe.

4

Article 24 - Consent

Consent must be freely given and specific. It is the lawful basis most retail blockchain products rely on.

The Immutability Tension

A public blockchain cannot delete or change a past record - clashing directly with the Article 23 right to erasure and rectification.

Four Recognised Mitigations

Off-chain Storage

Keep personal data off-chain. Write only a one-way hash to the ledger. The personal record remains deletable.

Crypto-shredding

Delete the encryption key. Encrypted on-chain data becomes permanently unreadable without altering the ledger.

Pseudonymisation

Replace direct identifiers with tokens. The on-chain record can no longer be linked to a natural person without the separate token map.

Redactable Ledger

A chameleon or redactable ledger allows controlled deletion. Rare and technically advanced; not yet mainstream.



WORKED EXAMPLE

Bahraini Bank - Loyalty Programme on Blockchain

The bank keeps each customer's personal details in a standard off-chain database and writes only a one-way hash to the chain.

If erasure is requested

The bank deletes the off-chain record. The on-chain hash remains but can no longer be linked to any person.

What this does not solve

The transaction pattern is still permanently visible on-chain. The design must also minimise what is written in the first place.